

List of publications – Ing. Roman Svoboda, PhD.

Impacted Journals:

1. Volume relaxation of a-Se studied by mercury dilatometry. R. Svoboda, P. Pustková, J. Málek, J. Non-Cryst. Solids 352 (2006) 4793-4799.
2. Relaxation behavior of glassy selenium. R. Svoboda, P. Pustková, J. Málek, J. Phys. Chem. Sol. 68 (2007) 850-854.
3. Structural relaxation of polyvinyl acetate (PVAc). R. Svoboda, P. Pustková, J. Málek, Polymer 49 (2008) 3176-3185.
4. Volume and enthalpy relaxation of a-Se in the glass transition region. J. Málek, R. Svoboda, P. Pustková, P. Čičmanec, J. Non-Cryst. Solids 355 (2009) 264-272.
5. Heat capacity of NaNO₂. L. Kourková, R. Svoboda, G. Sádovská, V. Podzemná, A. Kohutová, Thermochemica Acta 491 (2009) 80-83.
6. Relaxation in Ge₂Se₉₈ and As₂Se₉₈ glasses. R. Svoboda, P. Pustková, J. Málek, J. Non-Cryst. Solids 356 (2010) 447-455.
7. Apparent activation energy of structural relaxation for Se₇₀Te₃₀ glass. R. Svoboda, P. Honcová, J. Málek, J. Non-Cryst. Solids 356 (2010) 165-168.
8. Influence of sample form and thermal history on relaxation response. P. Honcová, R. Svoboda, J. Málek, Thermochemica Acta 507-508 (2010) 71-76.
9. Enthalpic structural relaxation in Te-Se glassy system. R. Svoboda, P. Honcová, J. Málek, J. Non-Cryst. Solids 357 (2011) 2163-2169.
10. Crystallization kinetics in Se-Te glassy system. R. Svoboda, M. Krbal, J. Málek, J. Non-Cryst. Solids 357 (2011) 3123-3129.
11. Interpretation of crystallization kinetics results provided by DSC. R. Svoboda, J. Málek, Thermochemica Acta 526 (2011) 237-251.
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13. Enthalpic relaxation in Ge₂Sb₂Se₅ glass. R. Svoboda, P. Honcová, J. Málek, J. Non-Cryst. Solids 358 (2012) 804-809.
14. Thermodynamic properties, decomposition kinetics and reaction models of BCHMX and its formex bonded explosive. Q.-L. Yan, S. Zeman, R. Svoboda, A. Elbeih, R. Matyáš, Thermochemica Acta 547 (2012) 150-160.

15. Kinetic analysis of the thermal decomposition of Zn(II)-2-chlorobenzoate complex with caffeine. L. Findoráková, R. Svoboda, *Thermochimica Acta* 543 (2012) 113-117.
16. Electrical conductivity and crystallization kinetics in Te-Se glassy system. J. Barták, R. Svoboda, J. Málek, *J. Appl. Phys.* 111 (2012) art. no. 094908.
17. Extended study of crystallization kinetics for Se-Te glasses. R. Svoboda, J. Málek, *J. Therm. Anal. Cal.* 111 (2013) 161-171.
18. Applicability of Fraser-Suzuki function in kinetic analysis of complex processes. R. Svoboda, J. Málek, *J. Therm. Anal. Cal.* 111 (2013) 1045-1056.
19. Thermal behavior and decomposition kinetics of Formex-bonded explosives containing different cyclic nitramines. Q.-L. Yan, S. Zeman, J. Šelešovský, R. Svoboda, A. Elbeih, *J. Therm. Anal. Cal.* (2013) 1419-1430.
20. Crystallization kinetics of a-Se, part 1: Interpretation of kinetic functions. R. Svoboda, J. Málek, *J. Therm. Anal. Cal.* 114 (2013) 473-482.
21. The effect of crystal structure on the thermal initiation of CL-20 and its C4 bonded explosives (II): models for overlapped reactions and thermal stability. Q.-L. Yan, S. Zeman, R. Svoboda, A. Elbeih, J. Málek, *J. Therm. Anal. Cal.* 112 (2013) 837-849.
22. Kissinger equation versus glass transition phenomenology. R. Svoboda, P. Čičmanec, J. Málek, *J. Therm. Anal. Cal.* 114 (2013) 285-293.
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24. Enthalpy relaxation in Ge-Se glassy system. R. Svoboda, J. Málek, *J. Therm. Anal. Cal.* 113 (2013) 831-842.
25. Structural relaxation in Se-rich As-Se glasses. R. Svoboda, J. Málek, *J. Non-Cryst. Solids* 363 (2013) 89-95.
26. Relaxation processes in selenide glasses: Effect of characteristic structural entities. R. Svoboda, *Acta Materialia* 61 (2013) 4534-4541.
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33. Analysis of crystallization in $\text{Sb}_2\text{Se}_{98}$ composition. P. Honcová, P. Pilný, R. Svoboda, J. Shánělová, P. Košťál, J. Barták, J. Málek, J. Therm. Anal. Calorim. 116 (2014) 613 – 618.
34. Crystallization behavior of $\text{Ge}_{17}\text{Sb}_{23}\text{Se}_{60}$ thin films. R. Svoboda, J. Přikryl, J. Barták, M. Vlček, J. Málek, Philos. Mag. 94 (2014) 1301-1310.
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42. Crystallization kinetics of Se-Te thin films. R. Svoboda, J. Gutwirth, J. Málek, T. Wágner, Thin Solid Films 571 (2014) 121-126.
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44. Kinetic fragility of Se-rich chalcogenide glasses. R. Svoboda, J. Málek, J. Non-Cryst. Sol. 419 (2015) 39-44.
45. Thermal behavior of Se-rich $\text{Ge}_2\text{Sb}_2\text{Se}_{(5-y)}\text{Te}_y$ chalcogenide system. R. Svoboda, J. Málek, J. Alloys Compd. 627 (2015) 287-298.

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53. Evaluation of glass-stability criteria for chalcogenide glasses: Effect of experimental conditions. R. Svoboda, J. Málek, J. Non-Cryst. Sol. 413 (2015) 39-45.
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61. Thermal properties and viscous flow behavior of As_2Se_3 glass. Z. Zmrhalová, P. Pilný, R. Svoboda, J. Šánělová, J. Málek, J. Alloys Compd. 655 (2016) 220-228.
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70. Correlation of structural, thermo-kinetic and thermo-mechanical properties of the $\text{Ge}_{11}\text{Ga}_{11}\text{Te}_{78}$ glass. R. Svoboda, D. Stříteský, Z. Zmrhalová, D. Brandová, J. Málek, J. Non-Cryst. Sol. 445-446 (2016) 7-14.
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94. Crystallization behavior of $(\text{GeTe}_4)_x(\text{GaTe}_3)_{100-x}$ glasses for far-infrared optics applications. R. Svoboda, D. Brandová, J. Alloys. Compd. 770 (2019) 564-571.
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34. Thermal behavior of amorphous $\text{Sb}_x\text{Se}_{1-x}$. P. Honcová, R. Pilař, R. Svoboda, P. Košťál, J. Málek, 11th Conference on Calorimetry and Thermal Analysis, 9.-13.9. 2012, Zakopane/Poland (poster)
35. Structural relaxation in Se-based glasses. R. Svoboda, J. Málek, International Days of Materials Science 2012, 25.-26.9. 2012, Pardubice/Czech Republic (poster)
36. Is the original Kissinger equation obsolete today? R. Svoboda, J. Málek, 4th Czech-Hungarian-Polish-Slovak Thermoanalytical Conference, 24-27.6 2013, Pardubice/Czech Republic (poster)
37. Analysis of crystallization in $\text{Sb}_2\text{Se}_{98}$ glass. P. Honcová, P. Pilný, R. Svoboda, P. Košťál, J. Barták, J. Málek, 4th Czech-Hungarian-Polish-Slovak Thermoanalytical Conference, 24.-27.6. 2013 Pardubice/Czech Republic (poster)
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39. Involvement of characteristic molecular entities in structural relaxation description. R. Svoboda, J. Málek, 23rd International Congress on Glass - ICG 2013 Prague, 1 – 5. 7. 2013, Praha/Czech Republic (poster)
40. Structural relaxation and crystal growth kinetics in Se-Te glasses. J. Málek, R. Svoboda, J. Barták, P. Košťál, 23rd International Congress on Glass - ICG 2013 Prague, 1 – 5. 7. 2013, Praha/Czech Republic (poster)
41. Structural relaxation in Se-rich chalcogenide glasses. R. Svoboda, J. Málek, 7th International Discussion Meeting on Relaxation in Complex Systems, 21.-26.7. 2013, Barcelona/Spain (poster)
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43. Crystallization behavior of $\text{Ge}_{17}\text{Sb}_{23}\text{Se}_{60}$ thin films. R. Svoboda, J. Barták, J. Přikryl, J. Málek, European Phase Change and Optoelectronics Symposium – EPCOS 2013, 8 – 10. 9. 2013 Berlín/Germany (poster)
44. Amorphous/crystalline transition in Te doped $\text{Ge}_2\text{Sb}_2\text{Se}_5$ glass. R. Svoboda, J. Gutwirth, J. Málek, 12th International Conference on the Structure of Non-Crystalline Materials (NCM12), 7-12.7.2013, Riva del Garda/Italy (poster)

45. Current evaluation methodologies for structural relaxation measurements. R. Svoboda, IDMS Winter Seminar, ReAdMat, 13.2.2013, Pardubice/Czech Republic (oral presentation)
46. Crystallization in Sb-Se system. P. Honcová, R. Svoboda, P. Pilný, P. Košťál, J. Barták, J. Málek, 35. International Czech and Slovak Calorimetric Seminar, 27. – 30. 5. 2013, Třeboň/Czech Republic (oral presentation)
47. Crystal growth kinetics, structural relaxation and viscosity behavior in selenium supercooled liquid. J. Málek, J. Barták, P. Košťál, R. Svoboda, IDMS Autumn Seminar, ReAdMat, 24. - 25.9.2013, Pardubice/Czech Republic (oral presentation)
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49. Study of influence of heating rate and particle size on crystallized products of $\text{Ge}_2\text{Sb}_2\text{Se}_{5-x}\text{Te}_x$ glasses. J. Gutwirth, R. Svoboda, P. Bezdička, J. Málek, IDMS Autumn Seminar, ReAdMat, 24. - 25.9.2013, Pardubice/Czech Republic (poster)
50. Crystal growth kinetics in amorphous Se-Te system. J. Málek, J. Barták, R. Svoboda, The International Symposium on Non-Oxide and New Optical Glasses – ISNOG, 24. – 28.8.2014, Jeju/Republic of Korea (oral presentation)
51. Correlation between crystallization kinetics of thin films and bulk glasses. R. Svoboda, J. Gutwirth, J. Málek, T. Wágner, The International Symposium on Non-Oxide and New Optical Glasses – ISNOG, 24. – 28.8.2014, Jeju/Republic of Korea (poster)
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53. Assessment of zeolites acid properties – unsolved challenging task. A. Martinez de Yuso, R. Bulánek, R. Svoboda, IDMS Autumn Seminar, ReAdMat, 16. - 17.9.2014, Pardubice/Czech Republic (oral presentation)
54. Preparation and physic-chemical properties of $\text{Ge}_8\text{Bi}_2\text{Te}_{11}$ and $\text{Ge}_8\text{Sb}_2\text{Te}_{11}$ thin films. V. Karabyn, J. Příkryl, L. Střížík, L. Beneš, R. Svoboda, B. Frumarová, M. Frumar, IDMS Autumn Seminar, ReAdMat, 16. - 17.9.2014, Pardubice/Czech Republic (poster)
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57. Influence of particle size on crystallization and relaxation behavior of $\text{Ge}_{20}\text{Se}_4\text{Te}_{76}$ glass for infrared optics. D. Brandová, R. Svoboda, Thermoanalytic seminar – TAS 2015, 18.6.2015, Pardubice/Czech Republic (poster)
58. Influence of experimental conditions on evaluation of glass-stability criteria. R. Svoboda, D. Brandová, J. Málek, Thermoanalytic seminar – TAS 2015, 18.6.2015, Pardubice/Czech Republic (poster)
59. Glass transition kinetics versus TA&C researchers. R. Svoboda, 3rd Central and Eastern European Conference on Thermal Analysis and Calorimetry, 25. - 28.8.2015, Ljubljana/Slovenia (invited lecture)
60. Crystallization from mechanically induced defects. R. Svoboda, D. Brandová, J. Málek, 3rd Central and Eastern European Conference on Thermal Analysis and Calorimetry, 25. - 28.8.2015, Ljubljana/Slovenia (poster)
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64. Kinetic analysis of complex processes – our Sleeping Beauty. R. Svoboda, 16th International Congress on Thermal Analysis and Calorimetry, 14.-19.8. 2016, Orlando/USA (invited oral presentation)
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67. Mechanically induced defects – origin of glass crystallization kinetic complexity? R. Svoboda, Slovak and Czech Glass Conference & Seminar on Defects in Glass, 28. – 30.6.2017, Trenčianské Teplice/Slovakia (poster).
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70. DSC baseline interpolations: Importance for kinetic evaluations. D. Brandová, R. Svoboda, 4th Central and Eastern European Conference on Thermal Analysis and Calorimetry, 28. – 31.8.2017, Chisinau/Moldova (lecture).
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73. Recrystallization kinetics of Enzalutamide with regard to processing, storage and handling conditions. J. Romanová, R. Svoboda, I. Obádalová, T. Pekárek, L. Krejčík, A. Komersová, 13th International Conference on Solid State Chemistry, 16.-21.9. 2018, Pardubice/ Czech Republic (oral presentation)
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77. Crystallization kinetics of Ge-Ga-Te far-infrared glasses. D. Brandová, R. Svoboda, The International Symposium on Non-Oxide and New Optical Glasses – 21st ISNOG, 17. – 21.6.2018, Quebec/Canada (poster)
78. Chemical interaction in solid state: Changes in behavior in matrix tablets due to the chemical interactions of individual components. J. Romanová, A. Komersová, R. Svoboda, T. Pekárek. 7th International Conference of Chemical Technology, 15. – 17.4.2019, Mikulov/Czech Republic (oral presentation)
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